



## THE THREE CADENCES OF PHYSICAL AI

Asynchronous Multi-Rate Temporal Hierarchy: Intent  $\rightarrow$  Action Chunks  $\rightarrow$  Microsecond Reflexes

### SYSTEM 2: INTENT

**Slow Cadence:** 0.5–2 Hz (500–2000 ms)

**Role:** Open-vocabulary reasoning

**Model:** Multimodal VLM (7B–70B)

**Output:** Expiring Intent Lease  $\mathcal{L}_{\text{intent}}$

**Authority:** Untrusted Proposal

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**Failure Mode:** If VLM stalls or drops frames, lease expires harmlessly.

Lease

### SYSTEM 1.5: TRAJECTORY

**Medium Cadence:** 20–50 Hz (20–50 ms)

**Role:** Action Chunk unrolling

**Model:** Diffusion ACT / Policy

**Output:** Horizon of  $H$  waypoints

**Authority:** Candidate Trajectory

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**Advantage:** Amortizes neural compute delay across continuous execution.

Chunks

### SYSTEM 1: REFLEX

**Fast Cadence:** 1000 Hz ( $1.0 \text{ ms} \pm 5 \mu\text{s}$ )

**Role:** Control Barrier Enforcement

**Model:** Bare-Metal FreeRTOS CBF

**Output:** Permitted Gate PWM ( $u_t$ )

**Authority:** **SOLE PERMISSION**

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**Safety Invariant:** Deterministic emergency halt on any timing fault.